

# DiracX developments: directions

Alexandre F. Boyer · CERN

12th DIRAC(X) Users' Workshop · FZU Prague · 13–16 October 2026

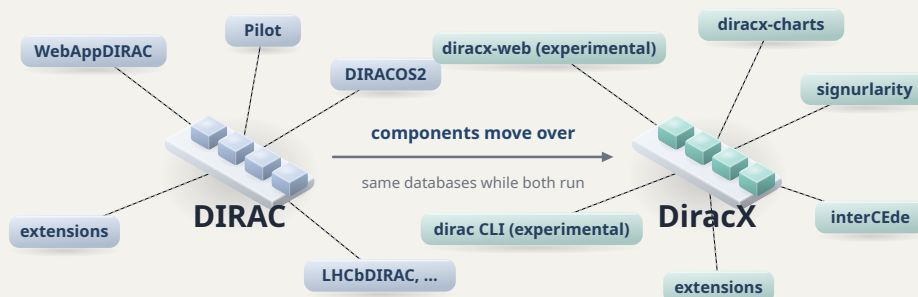


# | 1 • Where we are now

DIRAC and DiracX today



# Maintaining DIRAC, progressing on DiracX



**DIRAC** v9.1.21 · v9.0.27 · v8.0.85

WebAppDIRAC 6.0.6 · DIRACOS2 2.63

**Patched, not developed:** security fixes, and performance fixes when needed.

**DiracX** v0.6.0: foundations, authentication, jobs (search, status), sandboxes, read-only resource status, and experimental observability (OpenTelemetry).

The web app (diracx-web 0.1.0-a11) and the `dirac` CLI are still experimental.

**Adopt what already exists in DiracX.** DIRAC is only patched, so that DiracX can progress. LHCb runs one service and one task in production; two other installations are certifying DiracX.

## 2 · Evolving requirements, fruitful collaboration

New communities, new use cases, and two systems running side by side



# What DIRAC is for, and what it values

Running large-scale workflows of embarrassingly parallel jobs on heterogeneous, distributed computing and storage resources: high-throughput computing.

## Minimal dependencies

A database and a search index, on a few hosts.

**In DiracX:** standard protocols for what is external (S3, OpenTelemetry, OIDC). Without Kubernetes:

[Thursday's talk.](#)

## Multi-VO installations

One installation serves several communities.

**In DiracX:** multi-VO from the start.

## Extendable

Communities add their own pieces.

**In DiracX:** extensions, with a reference extension.

My own reading of what communities have asked for over the last few years.

# Requirements keep evolving

## New communities

**CMS and FCC**, who joined this summer, bring new requirements. CMS has no DIRAC installation and runs its own workload management: [Wednesday's CMS talk.](#)

## Advanced needs, in extensions

**CTAO, LHCb, Belle II and JUNO** built extensions for advanced needs that are now close to each other. They belong in the common project: CTAO's use of **CWL** turned out useful for DiracX as a whole.

## Constant needs

**GridPP** has few resources: DiracX must stay very simple to deploy, with minimal dependencies.

Share your needs early, at DOPs, before building around the system: someone else may need the same thing.

# What stays stable, what is new

## Stable

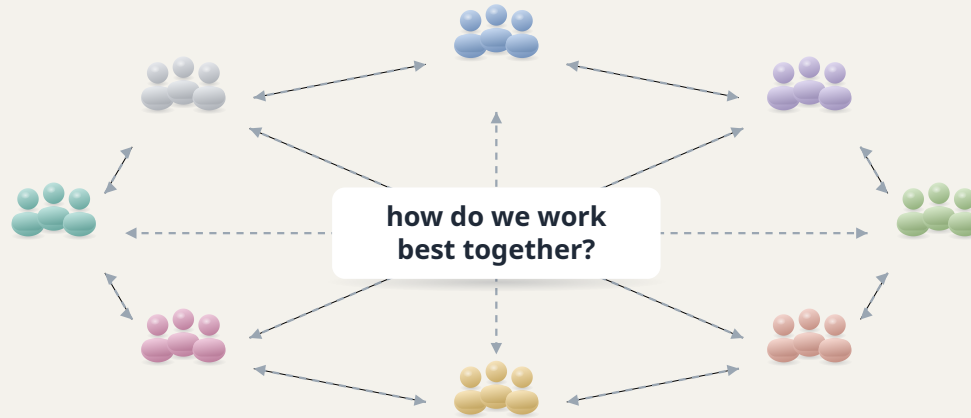
- **DIRAC keeps running**, with security and performance patches.
- Components move **one at a time**, over the same databases.
- **Extensions keep working** while components move.

## New

- **The transformation system**: a new data model, new logic, and a much richer scope than DIRAC's.
- **Decoupled from DIRAC**, so DIRAC's databases need no large changes and stay stable.
- **Standard protocols** (sandboxes in S3, observability in OpenTelemetry) and multi-VO from the start.

A stable path for what exists, and room for what is new.

# Many part-time hands



- **36 people** contributed since the last workshop, **10 to 18** in any given month, almost all part-time: about **5 to 6 full-time equivalents** in total, LHCb core developers included.
- They maintain DIRAC and develop DiracX at the same time.

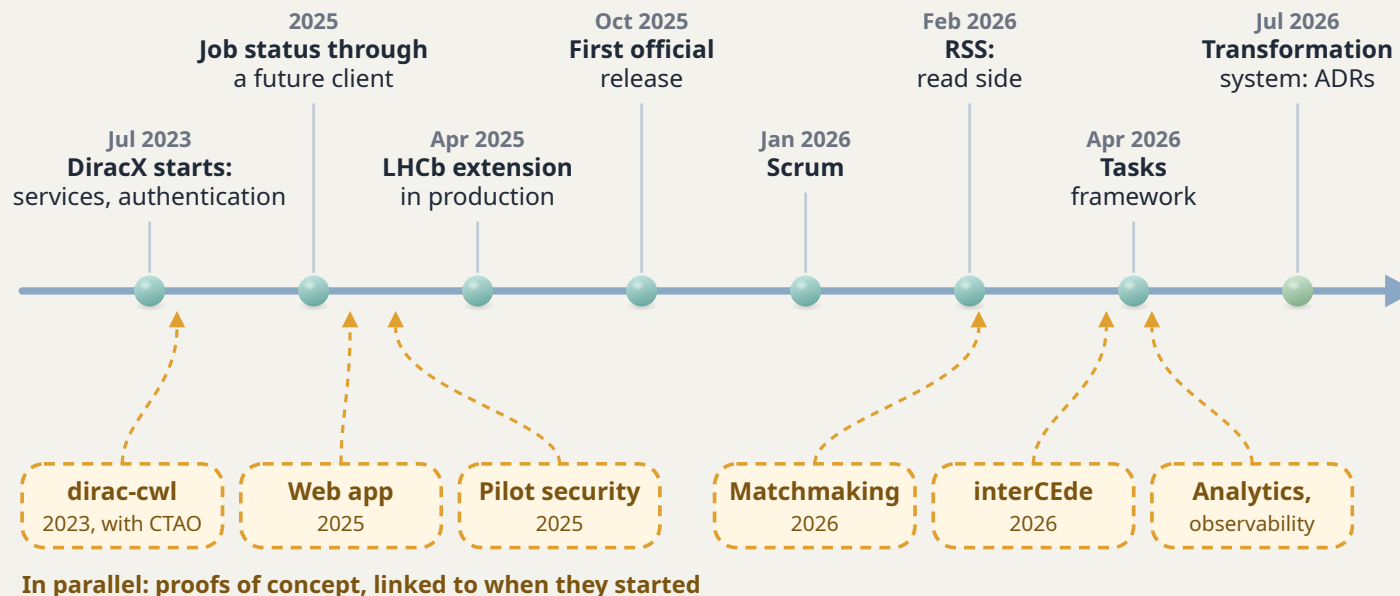
**Finding the best way to work with each other is paramount.**

## 3 · Adapting quickly

Responding to changing requirements, in small steps

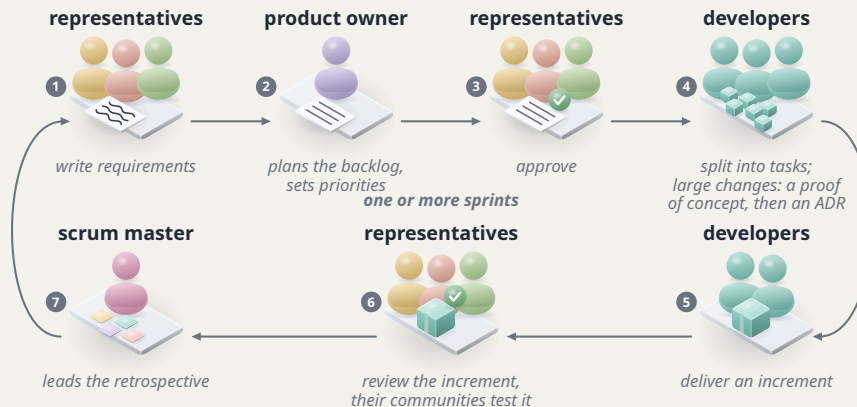


# The journey so far



- From 2023, three hackathons a year onboarded new contributors, with no dashboard or regular meeting to follow progress. Scrum brought both in 2026.
- Proofs of concept run in parallel, not mandatory: we try an idea before committing to it.

# Almost the current process

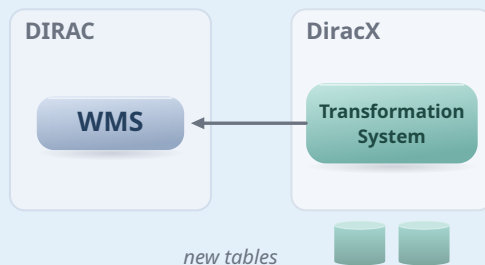


- **Scrum since 21 January 2026**: two-week sprints; planning and refinement in one weekly meeting; a monthly meeting with the representatives (DOPs); hackathons.
- **Large changes start with a proof of concept, then an ADR** that the other developers approve. ADRs and roadmap are public; the board is not.
- **We are moving towards this process; it is not there yet.** Like the roadmap, it adapts as we learn.

# Not a port: a reimplementation

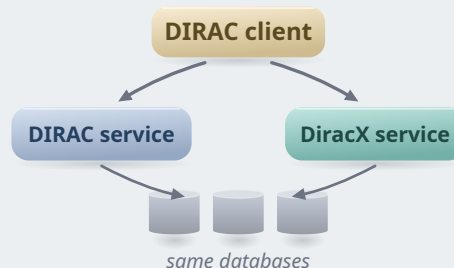
## A new data model

**The spine:** workgraph, transformations, parcels. New requirements need new tables and new logic. The RMS is used as it is, as a backend.



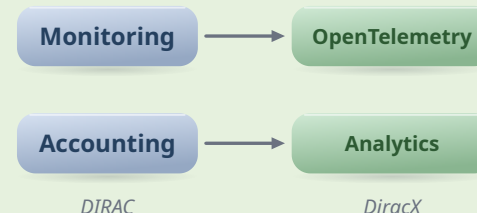
## Future clients

**DMS:** a DIRAC client calls DiracX instead of DIRAC, over the same databases. **RSS:** the read side is in DiracX; DIRAC's adapter is in review.



## Standard protocols

**Systems internal to DIRAC give way to standards:** monitoring to OpenTelemetry, accounting to analytics.



This week: [Data Management for DiracX](#) · [Analytics for DiracX](#) · [RSS in DIRAC and DiracX](#) (all Wednesday)

## | 4 · Results so far

Numbers since the last workshop



# Activity since the last workshop

| 17 SEP 2025 – 4 SEP 2026                | DIRAC STACK   | DIRACX STACK   | BOTH           |
|-----------------------------------------|---------------|----------------|----------------|
| Merged pull requests                    | 367           | 261            | 634            |
| People who merged one                   | 20            | 28             | 36             |
| People who gave a first review          | 12            | 11             | 17             |
| New contributors                        | 9             | 14             | 17             |
| Issues opened                           | 89            | 206            | 296            |
| Issues closed (completed / not planned) | 117 (76 / 41) | 224 (191 / 33) | 343 (268 / 75) |

DIRAC stack: DIRAC, WebAppDIRAC, DIRACOS2, Pilot. DiracX stack: diracx, diracx-charts, diracx-web, signurlarity, dirac-cwl, interCEde. Bots excluded; copies of one change on several release branches counted once. Issues to 2 Sep.

# Since Scrum: the backlog is shrinking

**295 → 195**

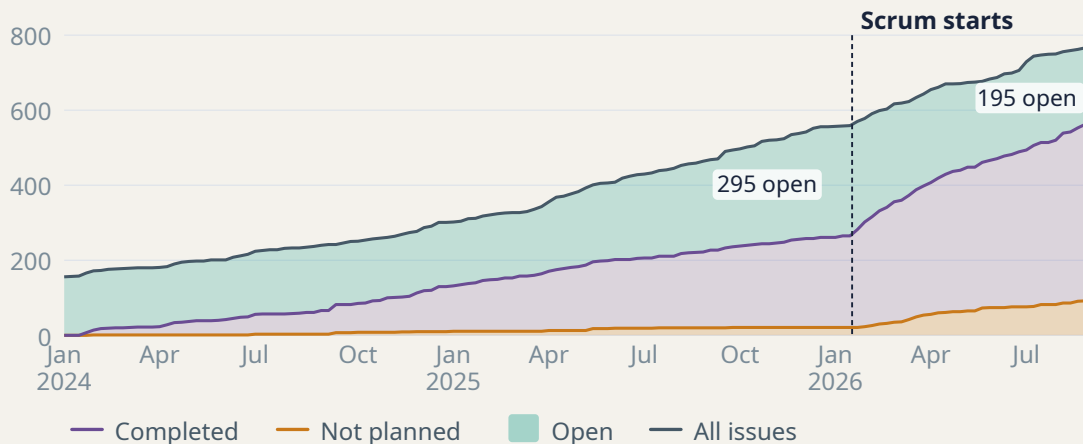
open issues, January 2026 to September

**10.6 → 36.6**

closures a month; 26.6 completed, 10.0 not planned or duplicate

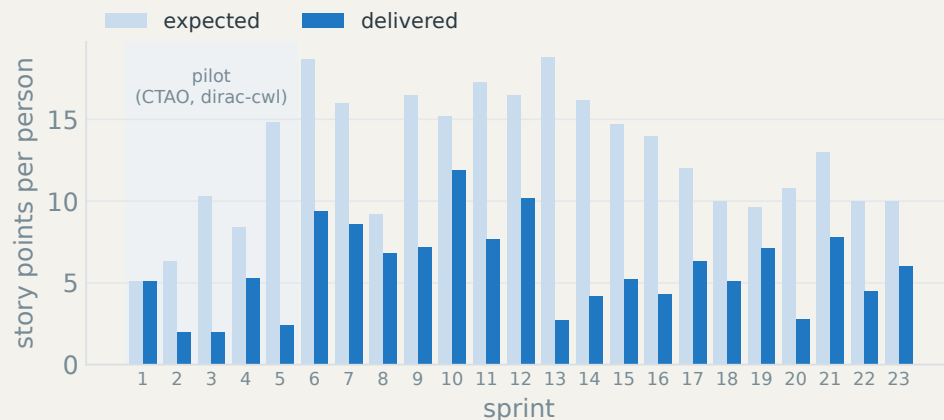
**16.3 → 27.3**

issues opened a month



All issues in the DIRACGrid repositories, bots excluded. Part of the fall is triage, and about 120 of the 195 still open are not triaged yet.

# Velocity



- **We expect about twice what we deliver:**  
since January, a median of 14 story points per person and sprint expected, 6.6 delivered.
- Delivered velocity follows availability:  
workshops, conferences, holidays.
- No sprint since January delivered what was expected; the best reached about three quarters.

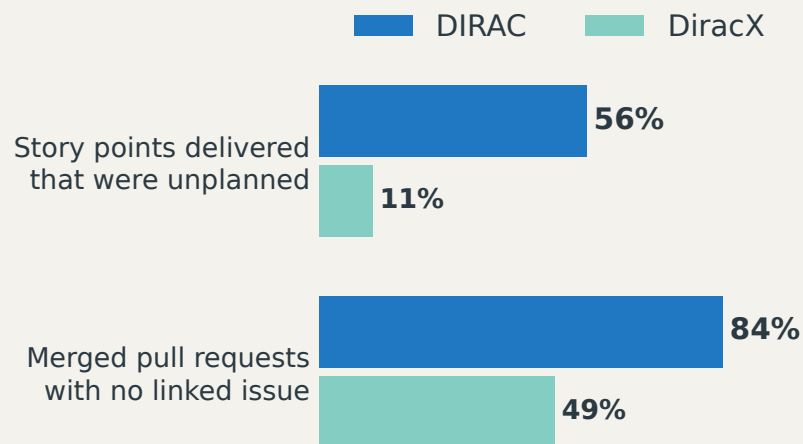
# Where an issue goes



*issues on the planning board, data to 4 September 2026*

- Issues that move on do so quickly. The ones still waiting have waited far longer.
- 121 issues sit in triage, most since the January board sweep. 34 are "in progress" with a median of two months.

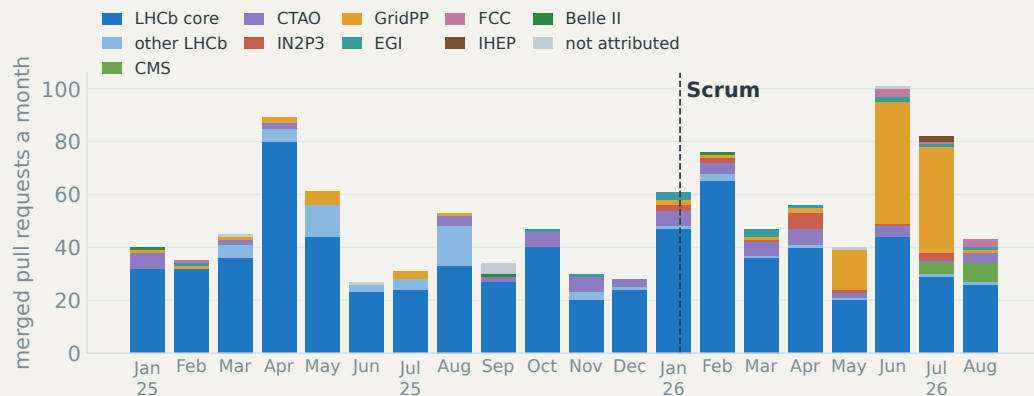
# Unplanned work lives in DIRAC



- Since January, **56% of the story points delivered on DIRAC were unplanned**, against 11% on DiracX.
- **84% of DIRAC pull requests have no linked issue**, against 49% on DiracX.
- Security and performance work arrives when it arrives. It is the cost of keeping DIRAC running, and why DIRAC only gets patches.

**For developers:** we overestimate what a sprint can hold, partly because we do not count the unplanned DIRAC maintenance. Plan for it.

# More hands, from more communities

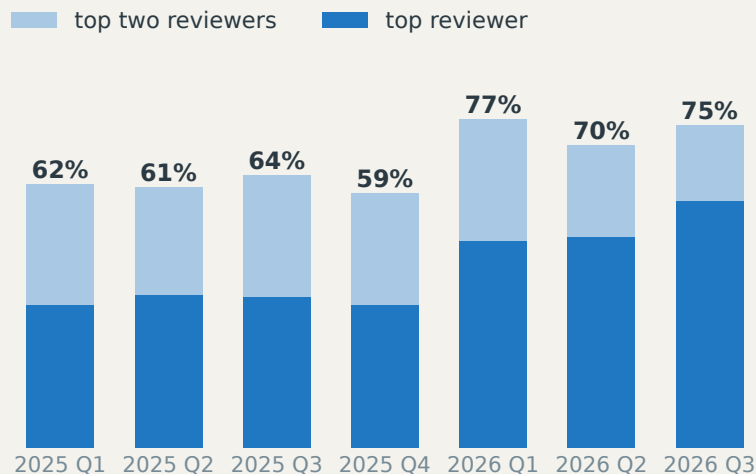


**36 people** from nine communities merged a change since the last workshop; **17 were new**. Almost all are part-time, so effort varies from month to month.

Outside the five core developers: **36% of merged PRs, 28% of issues opened, 53% of story points.**

Read with care: about half of the 2026 rise in PRs is **one GridPP developer** hardening DIRAC's SQL. CMS merged 14 changes since July.

# The bottleneck is review



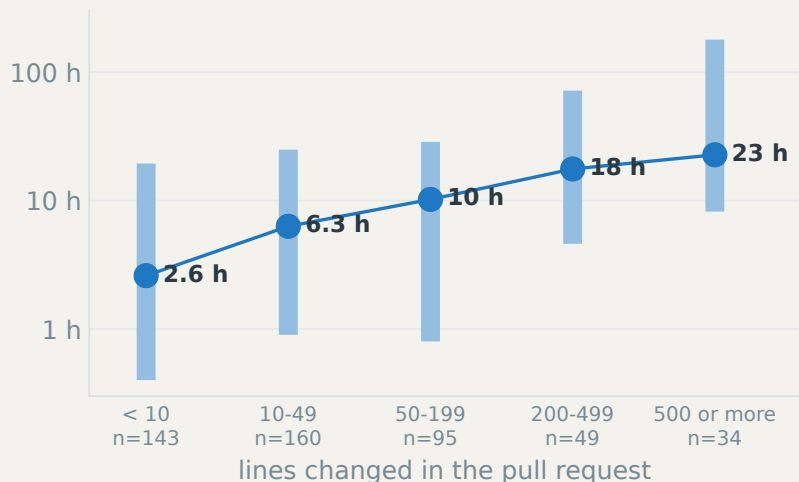
Share of first reviews, by quarter

- Two people give three quarters of first reviews. The top reviewer alone went from a third to over half.
- None of the 14 contributors new since January has given a first review yet.
- Design review is not in these numbers. ADRs are reviewed by everyone, in meetings and on chat.

→ *writing code has spread; reading it has not*

**Question for developers:** kanban style, should you review before starting something new once your tasks are done?

# Smaller pull requests get reviewed sooner



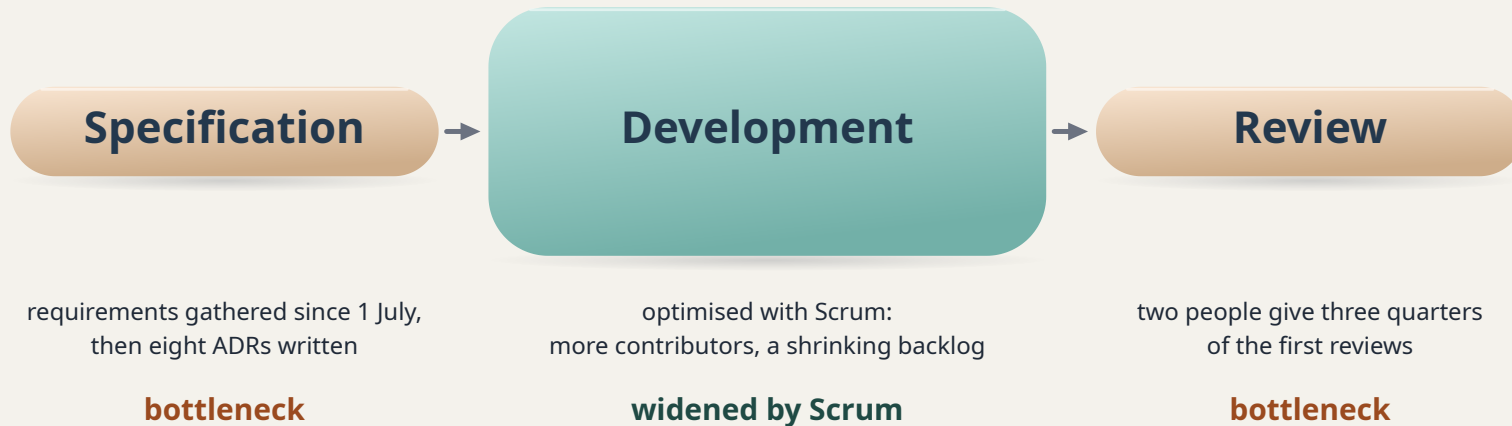
Median wait for a first review, with the middle half of the pull requests

- Under 10 lines: 2.6 hours. 500 lines or more: 23 hours. Time to merge goes from a few hours to five days.
- The spread is wide at every size, so size helps but does not decide.
- Who writes it matters too. For pull requests under 50 lines, the five core developers wait 2.5 hours; everyone else waits 11.

→ *split the work, and announce a big change early*

How to split a large change into stacked pull requests: [DiracX contributing guide](#)

# Where work waits



- **Scrum widened development:** more contributors, a shrinking backlog.
- **Specification and review are the narrow parts,** and they will matter more with every new component.

**For everyone in the room:** representatives, developers and users, where would you improve how we work?

# Cheaper code: easier to explore, same bottlenecks

"AI's primary role in software development is that of an amplifier."

[DORA 2025 State of AI-assisted Software Development](#)

- Writing code got cheap, so **spikes and proofs of concept** are easier to try: matchmaking, interCEde, analytics, observability.
- **Deciding what to build and checking it did** not get cheaper.
- It does **not change our process**. It shows its weak points sooner.

## Each one is something we just saw

- **Vague specification** → verbose, AI-generated issues that go stale (an open action). A clear ADR with a scenario test is what an agent, or a newcomer, can implement.
- **Review bottleneck** → more code means more to read, by the same two people.
- **Unplanned work** → faster code does not shrink urgent maintenance.
- **A new kind of user** → DiracX speaks HTTP and OAuth2, so an agent can use it through MCP.

→ *invest in specifications, tests and review, not in typing*

# | 5 · What's next

How we could improve, and the roadmap



# Next: building the transformation system

**Implementation starts at the hackathon this afternoon.** The technical details are in the next talk.

## Where it comes from

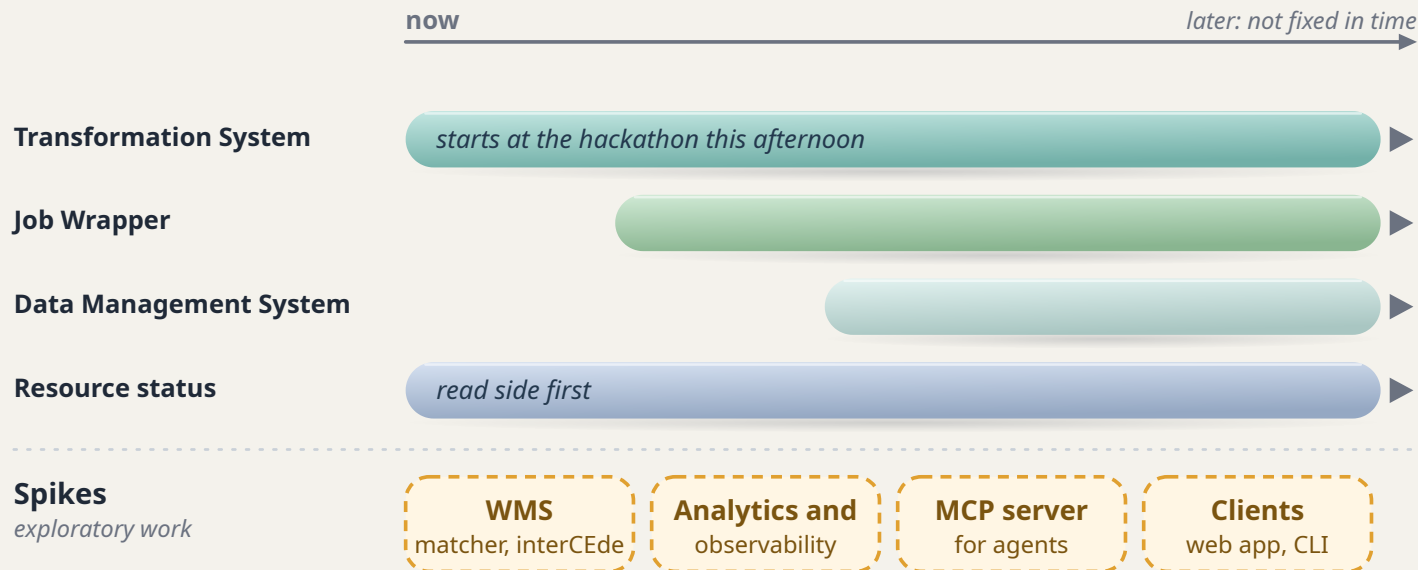
- **Use cases from the communities:** CMS, CTAO, FCC, Belle II and LHCb.
- **A design that comes from a proof of concept**, written down as ADRs.

## How it will be delivered

- **Increments the representatives can test**, shown at DOps whenever a sprint delivers one.
- **The more we are, and the better organised, the sooner.**

**This afternoon, two hackathons:** adapting your community's workflows to **CWL**, and building the **transformation system**.

# The roadmap moves with the communities



*Priorities, not commitments. New requirements can reorder this page, and the team writes the detailed plan.*

# If you run an installation today

## What you can count on

- **DIRAC keeps running**, with security patches and performance fixes when needed.
- **You choose when to switch each component**, usually at a shutdown.
- **No data migration** for anything but the transformation system, which has its own tables.
- **Your extension keeps working** on DIRAC while you move.
- **DIRAC and DiracX live in parallel**, and work can move gradually from one to the other.
- **Increments to test**, shown at DOpS whenever a sprint delivers one.

# Take-aways

---

## FOR INSTALLATIONS

- **Adopt what exists; DIRAC is only patched.** You move one component at a time, and the transformation system runs in parallel.

## FOR REPRESENTATIVES AND DEVELOPERS

- **Specification and review are the bottlenecks.** Increments you can test, shown at DOps, keep us close to your needs.

## FOR THIS AFTERNOON

- **Join one of the two hackathons:** adapting your workflows to CWL, or building the transformation system.

# Thank you

Questions and comments welcome, this week and after.

**Alexandre F. Boyer** · [alexandre.boyer@cern.ch](mailto:alexandre.boyer@cern.ch)

[github.com/DIRACGrid](https://github.com/DIRACGrid) · [diracx.diracgrid.org](https://diracx.diracgrid.org)

